End Term(Odd) Semester Examination December 2024

Rollno.....

Name of the Course and semester: B.Tech.CEand7th sem
Name of the Paper: Traffic Engineering and Management
Paper Code: TCE-713

Time: 3hours

MaximumMarks:100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a,b and c in each main question.
(iii) Total marks for each question is 20 (twenty). Each sub-question carries 10 marks.

Q1. (2X10=20Marks) CO1

a. Define traffic engineering and explain its importance in urban infrastructure.
b. A vehicle moving at 65 kmph speed was stopped by applying brake and the length of skid mark was 25.5 m. If the average skid resistance of the pavement surface is known to be 0.7. Determine the brake efficiency of the test vehicle. Also find out the retardation and time taken to stop the vehicle.
c. Discuss briefly road user characteristics and vehicular characteristics

Q2. (2X10=20Marks) CO2

a. Explain the significance of origin-destination surveys in traffic flow analysis.
b. An observer counts 360 vehicle/hr at a specific highway location. Assuming that arrival of vehicle at this highway location is Poisson distributed, estimate the probability of having 0, 1, 2, 3, 4, and 5 or more vehicles arriving over a 20 second time interval.
c. Explain how the speed and delay studies are carried out. What are the various uses of speed and delay studies?

Q3. (2X10=20Marks) CO3

a. Describe the principles of road markings and their importance in traffic management
b. Analyze the benefits and limitations of using street lighting for traffic safety at intersections.
c. The average normal flow of traffic on cross roads A and B during design period are 800 and 250 PCU per hour; the saturation flow values on these roads are estimated as 2500 and 2000 PCU per hour respectively. The all-red time required for pedestrian crossing is 12 seconds. Design two phase traffic signals by Webster's method.

Q4. (2X10=20Marks) CO4

a. Discuss the principles of rotary intersection design and their advantages over conventional intersections.
b. Explain grade separated intersections, the advantages and limitations.
c. What is Interchange? What are the different types of Interchange?

Q5. (2X10=20Marks) CO5

a. Define Transportation System Management (TSM) and its role in urban traffic management.
b. Discuss the impact of exclusive bus lanes on public transportation efficiency in metropolitan areas.
c. What is the need for traffic forecasting? Discuss forecasting method based on past trends and extrapolation.